

Carbon Emission, Income Inequality, and Redistribution in South Asia

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Abstract

The trade-off between carbon emission and inequality reducing policies is an issue of both discussion and contention. Traditional development plans in South Asia have been more focused on increasing the economy's growth rate, while environmental deterioration has taken a back place. Because of its physical and social environment, the region is particularly exposed to the horrors of climate change. including rising sea levels, melting Himalayan glaciers, and increasing storm frequency Despite these hurdles, continuous and rapid economic growth is required for the area to accomplish considerable poverty reduction, improve its people's economic well-being, and strengthen its resilience to environmental shocks and natural catastrophes, especially those caused by climate change. In this paper, we test whether there is a relationship between income inequality and carbon emission by using aggregate measures of these two variables. We find a positive and significant relationship between the two variables. We also find the evidence of existence of Environmental Kuznets Curve in South Asia. Lastly, we simulate an inequality

reducing policy

and see its effect on carbon emission and find out that a 20% decrease in inequality index leads to a 0.011% increase in carbon emission across South Asia. This trade-off between redistributive policies and carbon emission entails important policy implications.

Keywords: Inequality, Carbon emission, Redistribution

JEL: Q56, D63

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Certificate of Declaration

This is to certify that the research paper titled “Carbon Emission, Income Inequality, and Redistribution in South Asia” is our original research for the GRP South Asia initiative of SAESM. The paper’s indebtedness to other work and publication have been duly acknowledged.

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Introduction

Rapidly rising income inequality and global warming are two of the most pressing issues faced by the world today. While both these topics are hotly debated in isolation, the relationship between inequality and carbon emissions seldom features in general discourse.

In South Asia, conventional development strategies have been more focused towards escalating the growth rate of the economy, while the accompanying environmental degradation has taken a backseat. However, this idea of “grow now, clean later” can no longer be sustained.

The region is particularly vulnerable to the atrocities of climate change because of its geographical and socioeconomic landscape, including sea level rise, melting Himalayan glaciers, and increased frequency of typhoons. Notwithstanding these challenges, sustained and rapid economic growth is necessary for the region to achieve significant poverty reduction, uplift the economic well-being of its people, and increase its resilience to environmental shocks and natural disasters, including those associated with climate change. Rapidly increasing emissions threatens the growth, livelihoods, productivity, and well-being across all countries here.

Rising sea levels from global warming are eroding arable land in low-elevation coastal zones, posing a severe risk for rural incomes, food security, and commodity exports. Extreme weather events are responsible for more and more deaths each year. This not only has consequences for the economy but can also lead to social discontent. Thus, there is a pressing need to look towards more sustainable strategies for future growth.

The British charity Oxfam reported that the richest 10 per cent in the world contribute about 50 per cent of global carbon emissions, while the poorest 50 per cent contribute only 10 per cent (Oxfam International 2015). This implies that income inequality and emissions levels are inherently correlated. So, studying these two factors together and determining how they depend on each other has very important policy implications.

This cross-country study investigates the relationship between income inequality and carbon emissions at the regional level of South Asia for the period of 1980-2018. We also look at the effect of inequality reducing redistribution policies by simulating its effect on income inequality and carbon emission.

Literature Review

This section of the paper thematically analyzes the various works that provide a holistic secondary review covering the growth-inequality-environment and carbon emissions nexus. The paucity of data on South Asia has also rendered it crucial that we study specific case examples from both the developed and the developing economies for a thorough review and understanding of the ideas that have already been established, on both the sides of our hypothesis.

Analyzing the role of inequality for emissions is critical to understanding possible trade offs between pro-poor growth and climate change. However, much of the literature focuses on high-income countries. The findings of Liu et al. (2019) suggest that in countries like the US, the effects of income inequality depend on the time period in consideration. Higher income inequality increases carbon emissions in the short term, while it promotes carbon reduction in the long term. In 2000, Ravallion et al suggested in their paper that a trade-off exists between raising the average standard of living of people and controlling global warming. Similarly, Rojas-Vallejo's findings support the hypothesis that there is a trade-off between carbon emissions per-capita and income inequality. This trade-off is not homogeneous across countries and depends upon the level of developed measure by income per capita.

Although previous works of literature have shown the numerous paths in which income inequality can affect carbon emissions, Ravallion et al were the first to emphasize the importance of, and subsequently included, inequality within a country as a variable in this discussion along with the cross-country inequality variable. The authors have also broadened the discussion on inequality and carbon emission by showing that there exists a static trade-off between climate control and social equity. Their results suggest that - initially or in the short term - poverty reduction through simple redistribution of income or increasing the average income of the poor households result in more carbon-intensive consumption hence carbon emission increases. However, in the **long run**, a more pro-poor growth or a growth coupled with equity can yield a

desirable long-term trajectory of carbon emission. Moreover, by using a panel smooth transition regression (PSTR) estimation method developed by Gonzalez et al. (2017), it has been claimed that as income per-capita increases, the elasticity of inequality on carbon emissions increases monotonically from a negative value to slightly above zero. This suggests that the consumption effect decreases as income levels rise.

In addition to this, Ravallion et al. (2000) and Heerink et al. (2001) develop the consumption theory. They cite evidence of a non-linear relationship between income and environmental degradation at the household level. For example, Cropper and Griffiths (1994) find that at low levels of income, demand for firewood increases, causing deforestation. However, at higher levels of income the demand for firewood decreases as more modern forms of energy can be used. Depending on that income threshold different behaviors could be observed. Furthermore, Holtz-Eakin and Selden (1995) provide evidence that carbon emissions and income exhibits a positive but concave relationship. If this relationship is indeed concave at the household level, this theory predicts a negative relationship between inequality and emissions for a given level of income. Both Ravallion et al. (2000) and Heerink et al. (2001) test this theory and find this negative relationship, indicating that there is a trade-off between reducing inequality and reducing carbon emissions.

Grunewald et al. (2011) use Gini coefficients and emission data for forty eight years across multiple countries. Similar to Liu et al., they also find that for high-income countries with high income inequality, pro-poor growth and reduced per capita emissions levels go hand in hand. But for less inequal societies, reductions in income inequality are associated with increases in carbon emissions per capita. This thus, establishes that not only income, but also its distribution is an important factor that determines the level of aggregate carbon emissions and hence, the quality of the environment.

The literature review has also revealed that the measure of inequality that is used can cause a variance in trends. The literature covers the discussion on unique inequality measurements in order to explore different analytical approaches that link carbon emissions and income inequality. Quite interestingly, Hailemariam et al. (2019) investigate emissions in OECD countries, and find differing results depending upon the index of inequality used. An increase in the Gini index of inequality is associated with a decrease in carbon emissions. However, measuring top income inequality as the share of income earned by the richest 10% of the population, they find it to be positively correlated with carbon emissions. There is also some evidence in favor of the Environmental Kuznets Curve (EKC) in their study. Moreover, Grunewald et al. (2011) document a U-shaped relation between income inequality and carbon dioxide emissions per capita while the findings of Hailemariam et al. (2019) reveal a nonlinear relationship between economic growth and emissions as well. In the same vein, a recent study by Jorgenson et al tries to depict the changes in the CO₂ emissions at a sub-national level, using two measures of income inequality- the income share of top the 10% of population and Gini coefficients.

Coondoo and Dinda (2008) find evidence of heterogeneity across countries, showing that income inequality is not significant in a model that includes all countries, but becomes significant when countries are split into geographic groups. Likewise, Jorgenson et al. (2016) and Grunewald

et al. (2017) using imputed income inequality data from the Standardized World Income Inequality Database developed by Solt (2009) find mixed results for countries with different levels of income.

Marsiliani and Renstrom (2003) use sulfur, nitrogen and carbon dioxide and a different inequality measure: ratio of households ranked at top 90th percentile to the median household and two panels of 7 and 10 industrialized countries over 1978 –97. They argue that higher inequality leads to less environmental protection and consequently higher emissions in an overlapping generation's model with a majority elected representative. Moreover, their paper considers the inequality-emissions relationship for small groups of industrialized countries. Borghesi (2006) applies OLS and fixed effects panel data estimators to a panel data set of 37 countries for the period 1988 to 1995. He prefers the fixed effects estimator and finds that inequality does not have an effect on emissions. For the pooled OLS estimator, he finds a statistically significant negative effect between income inequality and carbon dioxide emissions. Heerink et al. (2001) use a cross section design with 65 country observations from about 1985. The specification includes income, its square, and the Gini coefficient. Similar to Ravallion et al. (2000), they find that income inequality is negatively associated with carbon emissions per capita.

Ravallion et al. (2000) use a pooled OLS model that interacts inequality with a third-order polynomial of income, a time trend, and population size. The panel data set consists of 42 countries over the period from 1975 to 1992. The authors use one (average) inequality measure per country. They find that there is a static tradeoff between reducing carbon emissions and reducing income inequality. All three studies [Ravallion et al. (2000), Borghesi (2006), Heerink et al. (2001)] rely on the inequality measure from the data set described in Deininger and Squire (1996).

Drabo (2011) finds, in a 2SLS fixed effects framework using 86 countries that inequality increases emissions. Gassebner et al. (2008) use extreme bound analysis for a sample of up to 120 countries in 1960–2001 and find that inequality is robustly associated with smaller CO₂ emissions. In the same line of reasoning as Boyce (1994), Borghesi (2006) suggests that an increase in inequality hinders the way for public policy solutions to environmental problems and therefore, greater inequality can contribute to increasing emissions.

Furthermore, the null effects of the Gini coefficient are generally inconsistent with the marginal propensity to emit approach. The approach suggests that as incomes become more evenly distributed, the poor will increase their consumption of energy and other products as they move into the middle class, resulting in an overall increase in anthropogenic emissions. The author believes policy approaches that combine both the equality enhancing effects with the direct reductions in emissions can be promising ways to curb carbon emissions to a notable degree, an idea worth exploring owing to its two pronged beneficial approach. In terms of specific examples,

the data seems to be ambiguous such that no generalizations can be made. For example, at the country level, Hao et al. (2016) study the case of China finding that as the income gap increases, carbon emissions per-capita also increases. Uzar and Eyuboglu (2019) find the same relationship for Turkey – reducing income inequality may reduce environmental problems. Baek and Gweisah (2013) and Jorgenson et al. (2017) address this question for the United States. Baek and Gweisah

(2013) find a positive association between income inequality and environmental degradation using Gini coefficients, while Jorgenson et al. (2017) fail to find statistical significance with the same measure of inequality.

In 1998, Torras and Boyce proposed the political economy theory for how income inequality can have a direct impact on environmental quality. They hypothesize that reducing income inequality will cause most people to demand higher environmental quality. Environmental quality is generally considered to be a normal good, meaning that consumers will demand more of it as their income increases. Those who experience a direct financial benefit from pollution

producing activities may not demand higher environmental quality as their income rises, but they are assumed to be a small group in this setup. Another way to think of this is that higher levels of inequality lead to pro-growth reforms that do not necessarily consider environmental degradation. Regardless of the interpretation, the political economy mechanism predicts a positive relationship between inequality and emissions. Torras and Boyce (1998) test their theory and find some supporting evidence for local pollutants such as sulfur dioxide.

The study by Hailemariam et al. mentioned above is specifically insightful as its measurement of the income share of the top 10% serves the greater purpose of analyzing political economy and how it can perpetuate and reinforce existing inequality (Boyce 1994) and Veblen effects (Veblen 1934). These analytical approaches show the potential economic and political power of the wealthy where they accrue resources by higher emissions and use it to survive as beneficiaries and the Veblen effect is when heightened consumption occurs due to emulative consumption of the wealthy. While the results of this longitudinal study show a positive association of carbon emission with income concentration of top 10%, the association with Gini coefficient remains insignificant.

Data

Compared to the conventional papers that the issues of inequality, this paper models a different measure of inequality than the GINI ratio. Because of the paucity of the GINI ratio for the majority of the South Asian countries, we use a ratio of the income share of the top 10 percent and bottom 50 percent population of the respective South Asian countries. It measures the average income difference between the poorest half and the highest earners within a population. Accordingly, higher the ratio, higher the inequality. Similarly, lower the ratio, lower the inequality.

$$\frac{\text{Income share of top 10\%}}{\text{Income share of bottom 50\%}} = \frac{10\%}{50\%} \times \frac{h}{h}$$

For the simulation part, the income inequality variable $\frac{\text{Income share of top 10\%}}{\text{Income share of bottom 50\%}}$ has been decreased by 20% in order to mimic an inequality reducing redistributive policy. The data on income share of the top 10 percent of the population and bottom 50 percent of the population is taken from World Inequality Lab Database. Similarly, the data of real GDP per capita at chained PPP of 2017 US\$

and

population is taken from Penn World Table 10.0. Moreover, data for carbon emission has been taken from Oak Ridge National Laboratory Database.

Empirical Strategy

Our empirical analysis is directly influenced by a much simpler version of Ravallion et. al. (2000)'s theoretical model that shows the dependence of aggregate carbon emission on the distribution of income as well as its mean.

We here assume that consumption of all the goods in an economy entails carbon emission either directly via consumption or indirectly via production. We say that each individual has an implicit derived demand function for carbon emission which is a function of income; such that $f(Y_{ij})$ is the derived demand for emission by person i in country j , with income Y_{ij} . The first derivative of this function is the marginal propensity to emit (MPE). The aggregate level of emission is assumed to be a simple sum of individual demand for emission. The average emission rate in a country j is

$$\bar{e}_{jt} = \frac{1}{N_{jt}} \sum_{i=1}^{N_{jt}} f(Y_{ijt})$$

With accordance to the basic principles of microeconomics, we assume that the derived demand for emission rises with a rise in income ($f'(Y_{ij}) > 0$) and that the MPE falls as income rises, ($f''(Y_{ij}) < 0$). From here, we can postulate that the aggregate rate of emissions increases with inequality reducing redistributive policies. We test whether this theory stands in the case of South Asian countries.

Our econometric model of emission rates is motivated by the argument given above. With compliance to our argument, we want a specification which allows for our third derivative to be non-zero. An extensive review of literature suggests that a log quadratic model where the log per capita carbon emission is regressed on log quadratic GDP per capita with inequality variable fits the data and the theory better and is parsimonious. Our central model takes the following form

$$\ln e_{jt} = \beta_1 + \beta_2 \ln y_{jt} + \beta_3 \ln^2 y_{jt} + \beta_4 I_{jt} + \beta_5 \ln y_{jt} I_{jt} + \delta_{jt} + \eta_{jt} \quad (1)$$

where, $\ln e_{jt}$ is the log per capita carbon emission of country j at year t , $\ln y_{jt}$ is GDP per capita of country j in year t , I_{jt} is the inequality index of country j in year t , N_{jt} is the population of country j in year t , t is the time trend while η_{jt} is the idiosyncratic error.

We estimate equation (1) using simple pooled OLS model and fixed-effects model in which country dummy variables are added. Then, we run a Hausman specific test to find out more valid estimates between the two.

Results

Table 1 gives our estimates of the pooled regression. Here, we can see that there is a significant

positive relationship between log carbon emission rate per capita. Other things being equal, with a 1% increase in log GDP per capita, the log carbon emission per capita increases by 0.64%. Similarly, here we can see that there is a significant negative relationship between squared log GDP per capita and log carbon emission per capita showing a non-linear relationship between two

variables. We also see that there is a significant positive relationship between log carbon emission per capita and log population size. With a 1% increase in population, the carbon emission per capita grows by 0.352%. Our main variable of concern, inequality has a significant positive relation with log carbon emission per capita. With 1% increase in inequality, we see that there is a 0.043% increase in carbon emission per capita. When other variables are interacted with the inequality measure, we don't see any significant results.

Table 1: Pooled OLS estimates

	Pooled OLS estimates
Log GDP per capita	0.6432* (0.2345)
Log GDP per capita interacted with inequality measure	0.0183 (0.0132)
Squared log GDP per capita	-0.2134* (0.076)
Squared log GDP per capita interacted with inequality measure	0.0021 (0.0064)
Log population size	0.352* (0.012)
Log population size interacted with inequality measure	-0.073* (0.0018)
Time trend	0.009 (0.006)
Time trend interacted with inequality measure	0.0012 (0.0008)
Inequality measure	0.043* (0.008)
Constant term	-3.276*

	(0.864)
R ²	0.786

Note: The dependent variable is the log of carbon emission rate per capita. There are 1092 observations. The error terms are Huber-White standard errors inside parentheses.

In our fixed effect model estimates in Table 2, we see that the effect of log population size is larger. With 1% increase in population, the carbon emission rate per capita increases by 1.287%. The relationship between log GDP per capita and log carbon emission rate per capita is positive and significant which infers that a 1% increase in GDP per capita leads to a 0.3982% increase in carbon emission rate per capita. Our main variable of concern inequality has a positive and significant relationship with carbon emission per capita. With a percentage increase in inequality, carbon emission rate per capita increases by 0.063%.

Table 2: Fixed effects model

	Fixed Effect estimates
Log GDP per capita	0.3982*

	(0.112)
Log GDP per capita interacted with inequality measure	0.0003 (0.0002)
Squared log GDP per capita	-0.127 (0.114)
Squared log GDP per capita interacted with inequality measure	0.0018 (0.0017)
Log population size	1.287* (0.282)
Log population size interacted with inequality measure	-0.023* (0.008)
Time trend	0.004

	(0.0038)
Time trend interacted with inequality measure	0.00013 (0.00011)
Inequality measure	0.063* (0.012)
Constant term	-6.353* (0.894)
R ² (within)	0.614

To test the validity of the results and to infer which method gives us the best results, we conduct a Housman specification test. The p-value of the test is greater than our 5% threshold level ($p\text{-value} = 0.1432$) with chi-squared value of 4.56. Hence, we infer that our Pooled OLS method estimates are more consistent and valid.

Now, to simulate the effect of an inequality reducing redistributive policy, we feed our central regression model with inequality data 20% less than the initial value, keeping everything else same. Our assumption is that this shows us the effect of redistributive policy that aims in reducing inequality. Table 3 gives both the pooled OLS estimates and fixed effects estimates of our simulation model. In our pooled OLS estimates, we see that when inequality increases by 1%, the carbon emission rate per capita increases by 0.054%, which is lower than the initial inequality level. Other estimates don't differ much from our initial inference.

Table 3: Pooled OLS and Fixed effect estimates of simulation model

	Pooled OLS estimates	Fixed Effect estimates
Log GDP per capita	0.5632 * (0.2157)	0.3215* (0.105)
Log GDP per capita interacted with inequality measure	0.0164 (0.0156)	0.0001 (0.0001)
Squared log GDP per capita	-0.1876* (0.054)	-0.116 (0.112)
Squared log GDP per capita interacted with inequality measure	0.0034 (0.0057)	0.0012 (0.0012)
Log population size	0.321*	1.153*

	(0.011)	(0.164)
Log population size interacted with inequality measure	-0.067* (0.0013)	-0.018 (0.006)
Time trend	0.007 (0.005)	0.0036 (0.0033)
Time trend interacted with inequality measure	0.008 (0.0003)	0.0000 (0.0000)
Inequality measure	0.054* (0.012)	0.074* (0.015)
Constant term	-2.78 * (0.544)	-4.328* (0.776)
R ²	0.732	0.604

Note: The dependent variable is the log of carbon emission rate per capita. There are 1092 observations. The error terms are Huber-White standard errors inside parentheses.

While using the fixed effect estimation technique, we find similar results as our initial fixed effects estimates. The coefficient of the variable of our concern, inequality shows that as inequality grows by 1%, the carbon emission per capita grows by 0.074%.

To test the validity of the results and to infer which method gives us the best results, we conduct a Housman specification test. The p-value of the test is greater than our 5% threshold level (*p-value* = 0.1567) with chi-squared value of 4.23. Hence, we infer that our Pooled OLS method estimates are more consistent and valid.

Conclusion

We find that there is a non-linear relationship between carbon emission and between-inequality in South Asia, giving evidence for existence of Environment Kuznets Curve (EKC). We also see a positive significant relationship between within-inequality and carbon emission per capita. When inequality rises by 1%, carbon emission per capita also increases by 0.043 %. When we tested our ground theory that denotes a trade-off between inequality reducing policies and carbon emission, we see that when inequality index decreases by 20%, a 1% increase in inequality leads to a 0.054% increase in carbon emission which is a 0.011% increase. Hence, we can say that with reduction in inequality, there is an increase in carbon emission rate. Which shows a trade-off between carbon emission and inequality reducing policies. This entails a huge policy implication.

While reducing inequality is very important, decreasing carbon emission in order to mitigate the effects of climate change is equally important. Policymakers who focus on inequality reducing measures should always take this trade-off into account.

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